Department of Electrical Engineering
Indian Institute of Technology, Madras

EE 2005 Electrical Machines - Quiz II

Class: II B.Tech
Duration: 50 min.

Date: 19 March 2024
Max. Marks: 20

17/20

Instructions to Candidates

1. Answer ALL questions.
2. For questions 1–4, pick the most appropriate alternative(s) from among the options (a,b,c,d) and write it down. **More than one alternative may be correct** and marks will be given only if all the correct options (only) are listed, legibly.
3. Answers to questions 5–7 need to be answered in the space provided. Write the steps clearly and sequentially.
4. Only the text written directly below the questions (numbered sheets) will be considered for marking and NOT what you may write in the back sides of pages.
5. Final answers are to be written to four significant digits of precision.

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$P=\left(\frac{2\pi N}{60}\right)T \Rightarrow T=17.5\,Nm$

1. A DC Motor has nameplate details as follows: Armature voltage 220 V, Field Current 0.8 A, Speed 1200 RPM, and Power 2.2 kW. This means that (seperately excited)

   (a) the rated torque is 17.5 N m and rated field voltage is 220 V.

   (b) the rated torque is 1.8 N m and rated field voltage is 220 V.

   (c) the rated torque is 17.5 N m, but the field voltage depends on the field resistance.

   (d) the rated torque is 1.8 N m, but the field voltage depends on the field resistance.

   Ans.: The correct option(s) is/are: (c)

2. In a DC machine, the commutator-brush system is used to

   (a) convert the alternating armature induced EMF to DC as seen from brush terminals.

   (b) convert the dc induced EMF in armature to alternating voltage as seen from brush terminals.

   (c) convert the dc induced EMF in armature to dc voltage as seen from brush terminals.

   (d) convert the ac induced EMF in armature to ac voltage as seen from the brush terminals.

   Ans.: The correct option(s) is/are: (a)

3. A dc machine has an emf constant of 1.2 V/RPM. When a current of 10 A is flowing in the armature, the torque developed is

   $1.2\times 60/2\pi\times 10$

   (a) 120 N m (b) 12 N m (c) 114.5 N m (d) 1.2 N m

   Ans.: The correct option(s) is/are: (c)

4. When a dc motor having an emf constant of 0.1625 V/RPM and armature resistance of 0.5 Ω is running at a speed of 1200 RPM when supplied at 200 V, the armature current drawn is

   (a) 2.5 A (b) 5 A (c) 10 A (d) 15 A

   Ans.: The correct option(s) is/are: (c)

$200 = 0.1625\times 1200 + Ia(0.5)$

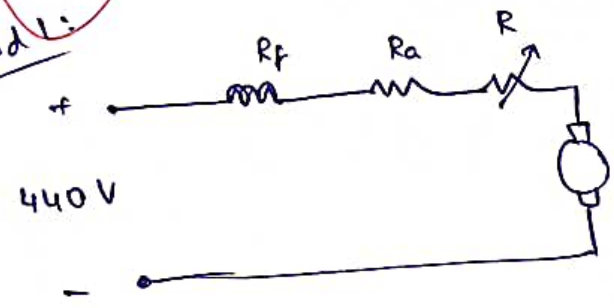
$440 = 110 \times R$

5. A 440 V series motor has a total series resistance (including field) of 0.8 Ω. It takes 110 A to give the rated output at 800 RPM. Determine the resistance to be added to obtain rated torque at starting. Assume that the flux is proportional to the current. (3 marks, binary marking)

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Ans.: The resistance to be added is: 3.2 Ω

Method 1:



$E = K_t I_a \omega$

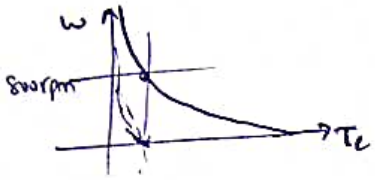
$$440 - 0.8 \times 110 = K_t \times 110 \times 800 \times \frac{2\pi}{60}$$

At starting, $\omega = 0$

$$(T_e) = K_t \cdot I_a^2 = K_t \cdot 110^2 \quad \text{(for rated torque)}$$

$$I_a' = \frac{440}{0.8 + R} = 110 \text{ A}$$

$$\Rightarrow \boxed{R = 3.2\ \Omega}$$



Method 2:

$$\omega = \frac{V}{\sqrt{K_t T_e}} - \frac{R_a + R_{sh}}{K_t}$$

$$800 = \frac{V}{\sqrt{K_t \cdot T_e}} - \frac{R_a + R_{sL}}{K_t}$$

$$0 = \frac{V}{\sqrt{K_t T_e}} - \frac{R_a + R_{sL} + R}{K_t}$$

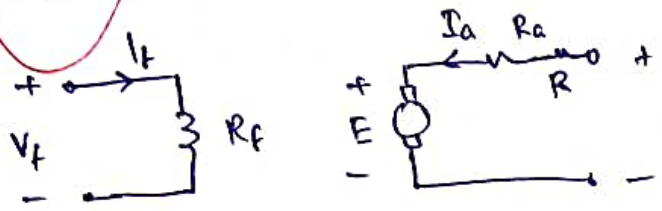
$$\Rightarrow \frac{R}{K_t} = 800 \times \frac{2\pi}{60}$$

$$R = K_t \times 800 \times \frac{2\pi}{60}$$

$$\Rightarrow \boxed{R = 3.2\ \Omega}$$

6. A separately excited dc motor rated for 220 V, 60 A operation, having armature resistance of 1 Ω, back-emf constant of 0.9 V sec/rad, is connected to a mechanical load having a characteristic $T_{load} = 0.0037\omega^2$, where ω is speed in rad/s. The source voltage is 220 V. Find the external resistance to be added in order to get a speed of 104 rad./s. (5 marks)

Ans.: The resistance to be added is: 1.843 Ω



$K_e = 0.9$ Vs/rad , ~~$I_a = 60$ A~~

$\Rightarrow$ ~~$220 - 60 \times 1 = E = 0.9 \times \omega = 160$ V~~

$\Rightarrow$ ~~$\omega = 177.78$ rad/s~~

Resistance R.

$T_{load} = T_e \Rightarrow 0.0037\omega^2 = 0.9 \times I_a$

$220 - I_a(1 + R) = 0.9 \cdot \omega$

~~$I_a = 220$~~ We want $\omega = 104$ rad/s

$\Rightarrow I_a = 44.465$ A $\Rightarrow 220 - 44.465(1 + R) = 93.6$

$\Rightarrow R = 1.843\ \Omega$

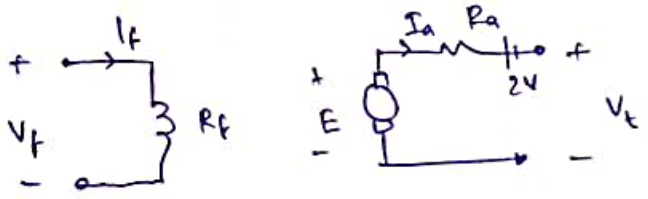
7. A dc machine has a brush drop of 2 V at rated current of 20 A. At the rated speed of 1200 RPM, the iron losses are 20 W and the induced emf is 190 V. The armature resistance is 0.5 Ω and the mechanical losses may be considered constant at 15 W. Find the efficiency of the machine if (a) this is operating in generator mode (b) this is operating in motor mode. The field being separately excited may be assumed to draw a constant power of 100 W. (8 marks)

Ans.: The efficiency in generator mode is: 90.47%

Ans.: The efficiency in motor mode is: 91.02%

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(a) Generator :



$I_a = 20A$, $E = 190\,V = k_e \times 1200 \times \frac{2\pi}{60}$

$V_t = 190 - 20 \times \frac{1}{2} - 2 = 180 - 2 = 178\ V$

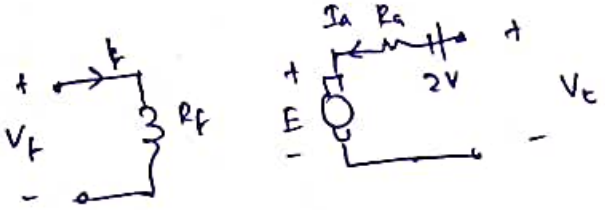
$P_{out} = 180 \times 20 = 3600 - 40 = 3560\ W$ (− 2×20)

$P_{in} = E_b \cdot I_a + P_{losses} = 190 \times 20 + 20\,W + 15\,W + 100\,W$

$= 3835\ W$

4 ⇒ $\eta = \frac{P_{out}}{P_{in}} \times 100\% =$ 90.47%

(b) Motor:



1 $V_t - 2 - R_a I_a = E = 190 = k_e \times 1200 \times \frac{2\pi}{60}$

⇒ $V_t = 190 + 2 + 20 \times \frac{1}{2} \overset{+5}{=} 202\ V$

$P_{in} = 202 \times 20 + 20 + 15 \overset{+100}{=} 4075\ W$

$P_{out} = E_b I_a = 190 \times 20 = 3800\ W$ ⇒ $\eta =$ 91.02%